

uart

How uart was designed and verified, with every number read from its recorded gate results. A draft: the block is not released.

What it is

A UART transmitter: one start bit, 8 data bits (LSB first), one even-parity bit, one stop bit, each held `CLKS_PER_BIT` clocks. Second rung on the /vde corpus ladder - harder than counter8, and the one the holdout fault ("UART parity inverted where the visible tests do not look") is written about.

Ports: `clk` input, `rst` input, `start` input, `data[7:0]` input, `tx` output, `busy` output. Clock period 10 ns (100 MHz).

REQUIREMENTS, FROM SPEC/SPEC.YAML

ID	CHECK	REQUIREMENT
REQ-IDLE	sim	While idle (no frame in flight), tx is 1 and busy is 0.
REQ-RESET	both	When rst is sampled high on a clock edge, the transmitter is idle on the next clock, with tx at 1 and busy at 0, regardless of any frame in flight.
REQ-LATCH-ON-START	sim	A one-clock start pulse while idle latches data on that clock, and later changes to data during the frame do not alter the bits sent.
REQ-START-BIT	sim	The first CLKS_PER_BIT clocks of a frame drive tx to 0 (the start bit).
REQ-DATA-LSB-FIRST	sim	After the start bit the frame sends data[0] through data[7] in that order, one bit per CLKS_PER_BIT clocks.
REQ-PARITY-EVEN	sim	The bit after data[7] is the even parity bit, equal to the XOR of all eight latched data bits, so the nine bits data[7:0] plus parity contain an even number of ones.
REQ-STOP-BIT	sim	The last CLKS_PER_BIT clocks of a frame drive tx to 1 (the stop bit), after which tx stays at 1 until the next frame's start bit.
REQ-BIT-HOLD	sim	Every bit of the frame is held on tx for exactly CLKS_PER_BIT consecutive clocks, so a frame is exactly 11 * CLKS_PER_BIT clocks long.
REQ-BUSY-FRAME	sim	busy rises with the start pulse and stays 1 until the stop bit's CLKS_PER_BIT-clock hold completes, then returns to 0.
REQ-START-WHILE-BUSY	formal	A start pulse while busy is 1 is ignored; the frame in flight is not disturbed and no second frame is queued.
REQ-TX-IDLE-HIGH-WHEN-NOT-BUSY	formal	Whenever busy is 0, tx is 1 (the line never drives a data or start bit outside a frame).

Architecture, and why

FIGURE 1 *block diagram*



Drawn from the ports in spec/spec.yaml.

The decisions recorded while the block was designed, each with its reason:

- **CLKS_PER_BIT default = 4** Brief leaves it open; 44-clock frame keeps sim/formal fast while the hold counter stays a real 2-bit counter so off-by-one faults are visible (P2)
- **rst wins over a same-clock start** Simplest reading of REQ-RESET: transmitter idle next clock, no frame begins (P2)
- **tx/busy registered; busy=1 for exactly 11*CLKS_PER_BIT clocks from the edge after start is sampled** Fixes the frame origin the tb and RTL both need; busy covers the full stop-bit hold and falls the clock after it completes (P2)
- **Clock period 10 ns, single domain clk** Brief gives no frequency; smallest round period where every requirement is whole clocks (P2)
- **sim wo-1 (routed rtl) given to tb-writer in work-order mode** rtl-writer diagnosis: tb Bench.clock reads dut.data.value right after writing it (cocotb deferred write), model latches stale byte; all 5 fails are data bits at frame clock>=4 (P4)
- **escalate mutate: 4 equivalent survivors, kill rate capped at 0.79 < 0.90** tb-writer rebuilt mutant 6 with mutate_runner and ran the suite (all pass, per-clock model check included); survivors unobservable at ports at CLKS_PER_BIT=4. SKILL.md: equivalent mutants are an escalation for a human (bounds waiver vs RTL restructure); no person present, so the rung stops at the fix loop. (P4)

How it was verified

4/14

gates with a fresh recorded pass

79.0%

mutants killed by the testbench

TABLE *the vde gates, as recorded*

GATE	CHECKS	RESULT	RECORDED NUMBERS
cover	code coverage	not run	no recorded result
drc	magic + klayout DRC	not run	no recorded result
formal	SymbiYosys proof	not run	no recorded result
glsim	gate-level sim	not run	no recorded result
harden	LibreLane to GDS	not run	no recorded result
holdout	held-out tests	pass	5/5 tests pass 2026-09-24 23:59
lint	Verilator lint	pass	0 findings, 0 failing 2026-09-24 23:58
lvs	netgen LVS	not run	no recorded result
mutate	mutation score	fail	15/19 killed; kill rate 79.0% 2026-09-25 00:00
precheck	Tiny Tapeout precheck	not run	no recorded result
sim	cocotb simulation	pass	6 tests pass 2026-09-25 00:04
spec_lint	spec is checkable	pass	0 findings, 0 failing 2026-09-24 23:45
synth	Yosys synthesis	not run	no recorded result
timing	OpenSTA signoff	not run	no recorded result

For `sim` the full result of the run on record was not kept, so the row gives only the numbers `state.json` recorded with it.

**WHAT WAS NOT
VERIFIED**

9 of the 14 gates this block owes have no recorded result, so this document has no numbers for them: cover, drc, formal, glsim, harden, lvs, precheck, synth, timing. Recorded as failing or stale: mutate.

Performance, power and area

Area: **no synth result is recorded**. Timing: **no timing result is recorded**. No gate in this flow measures power, so this document gives no power figure.

Layout

No layout render: no GDS in the workspace (the block has not been hardened or laid out).

Release record

Not released. no reports/checks.json. No human sign-off (H2) is recorded. 0 waivers recorded.